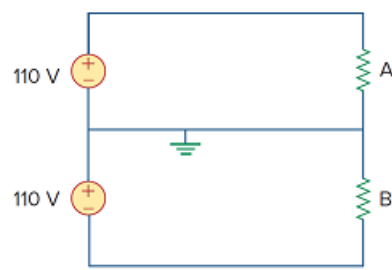


Problem

A three-wire system supplies two loads *A* and *B* as shown in Fig. 2.125. Load *A* consists of a motor drawing a current of 8 A, while load *B* is a PC drawing 2 A. Assuming 10 h/day of use for 365 days and 6 cents/kWh, calculate the annual energy cost of the system.

Figure 2.125:



Step-by-step solution

Step 1 of 3

Consider the following three-wire system:

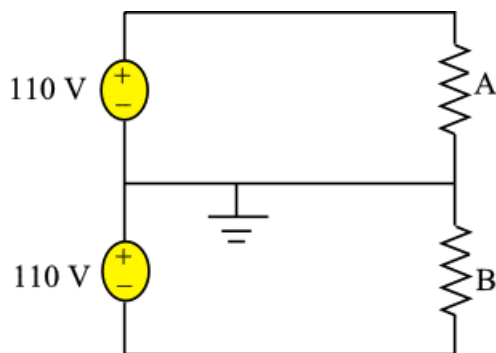


Figure 1

Step 2 of 3

Load *A* is a motor drawing a current of 8 A.

Load *B* is a PC drawing a current of 2 A.

The usage of electricity is 10 h/day for 365 days and the cost of electricity is 6 cents/kWh

The voltage across loads *A* and *B* is 110 V each.

Determine the power consumed by Load *A*.

$$\begin{aligned} P_A &= VI \\ &= (110)(8) \\ &= 880 \text{ W} \end{aligned}$$

Determine the power consumed by Load *B*.

$$\begin{aligned} P_B &= VI \\ &= (110)(2) \\ &= 220 \text{ W} \end{aligned}$$

Step 3 of 3

Determine the total power consumed by two loads.

$$\begin{aligned}P &= P_A + P_B \\&= 880 + 220 \\&= 1100 \text{ W} \\&= 1.1 \text{ kW}\end{aligned}$$

Determine the energy consumption in kWh for 365 days.

$$\begin{aligned}E &= P(10 \text{ h/day})(365 \text{ days}) \\&= (1.1 \text{ kW})(10 \text{ h/day})(365 \text{ days}) \\&= 4015 \text{ kWh}\end{aligned}$$

Determine the cost of energy.

$$\begin{aligned}C &= E(6 \text{ cents/kWh}) \\&= (4015)(6) \\&= 24,090 \text{ cents}\end{aligned}$$

Thus, the annual energy cost of the system is 24,090 cents.